

Global Considerations in Kidney Disease: Indian Subcontinent

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CHAPTER OUTLINE

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KEY POINTS

- Community-acquired acute kidney injury (AKI) secondary to tropical infections, use of indigenous drugs, toxic envenomation, and obstetric factors are some of the common causes of AKI in this region.
- Peritoneal dialysis in acute settings holds promise in resource-constrained and far-off settings, where it can be used with minimal expertise and infrastructure to treat life-threatening complications of kidney failure, especially AKI.
- Chronic kidney disease (CKD) is a major emerging noncommunicable disease in this region on account of the rapid rise in population and high prevalence of risk factors such as diabetes, hypertension, and nephrolithiasis.
- CKD of undetermined cause is common in this region and is mostly seen in young men, who are often in rural areas and work outdoors. The clinical presentation is usually late and suggests chronic tubulointerstitial involvement. It is likely that a host of environmental factors or toxins in conjunction with chronic dehydration play a role.
- Long-term renal replacement therapy is unavailable to most patients with end-stage kidney failure because of inaccessibility and unaffordability. However, joint public and private efforts have shown promise in establishing self-sustaining programs to provide therapy.

The Indian subcontinent occupies the southern portion of the Asian continent and includes India, Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan, and the Maldives. The subcontinent is home to well over 25% of the world's population, making it the most densely populated region in the world. Most of the population of the region lives in rural areas, relies on agriculture for its livelihood, and has limited access to health care. The countries of the subcontinent fall into the category of medium human development, as measured by the human development index.

India is the largest country in the subcontinent, with a population of more than 1.4 billion, and is also the largest economy in the region. It has enjoyed impressive economic growth, as a result of which India has been put into the category of "newly industrialized countries," a classification between industrialized and developing countries. The mean per-capita gross national product has grown, but World Bank data state that, as of 2019, 562 million people still continued to live on less than US\$2 a day. Economic and development indicators of major countries of the region are shown in

Table 77.1. This combination of underdevelopment, plus a rapidly growing economy, is reflected in the disease spectrum. On the one hand, countries of the subcontinent are challenged with epidemics of infectious disease, and on the other hand, diabetes mellitus, the so-called disease of the affluent, has reached epidemic proportions among poor people living in urban areas. Health care in the public sector in these countries is organized in the shape of a pyramid, with primary health centers at the bottom, followed by intermediate-level hospitals and referral hospitals at the top. Specialized care for disease is usually available only at major referral hospitals. There are only about 2600 nephrologists in India with one of the lowest nephrology workforce densities worldwide.¹ There is a thriving private-sector health care industry, but treatment costs are quite high and only the rich or patients whose expenses are covered by their employers can afford treatment in these hospitals.

The differences in living standards and availability of health care determine the variations in disease patterns, management practices, and disease outcomes. The spectrum

Table 77.1 Economic and Development Indicators on the Indian Subcontinent

Parameter	India	Pakistan	Bangladesh	Sri Lanka
Population (billions)	1.22	0.193	0.163	0.021
Population growth rate (%)	1.31	1.55	1.58	0.91
Life expectancy at birth (years)	67.48	66.7	70.36	76.15
Median age (years)	26.7	22.2	23.9	31.4
Percentage of total population that is elderly (>65 years)	5	4	5	8
Literacy rate (%)	64.84 (in 2011)	55 (in 2009)	57.7 (in 2011)	92.5 (in 2010)
Infant mortality rate/1000 live births (in 2013)	44.6	59.35	47.30	9.24
Per-capita gross domestic product purchasing power parity (US\$; in 2012)	3800	3100	2000	6000
Percentage of the population living below the national poverty line	29.8	22.3	31.5	8.9
Human development index	0.554	0.515	0.515	0.715
Country ranking by human development index (of 179 countries)	136	146	146	92

Data from The World Development Report 2022 <<https://data.worldbank.org/en/publication/wdr2022>>, Global Burden of Disease <<http://www.healthdata.org/gbd>>, World Health Organization Global Health Observatory <<http://www.who.int/gho/countries/en/>>, and CIA World Factbook <<https://www.cia.gov/library/publications/resources/the-world-factbook/>>

of kidney diseases in this region is characterized by a mix of conditions that are globally encountered and those specific to the subcontinent. The latter can be secondary to a genetic predisposition in specific ethnic groups or related to exposure to environmental factors, such as climatic conditions, infectious agents, envenomation, and chemical toxins. Genetic factors that predispose to kidney disease in the tropics include glucose-6-phosphate dehydrogenase (G6PD) deficiency, which gives rise to intravascular hemolysis and pigment-induced acute kidney injury (AKI). Indigenous health care systems are still popular in rural areas, and patients are frequently treated with herbs and potions that could add to the burden of kidney disease.

ACUTE KIDNEY INJURY

Reliable statistics on the patterns and prevalence of AKI in this region are unavailable, but it is the most encountered renal emergency associated with high mortality.² For example, about 0.1% to 0.25% of all admissions to the authors' hospital admissions are for AKI. Most of the causes of AKI described elsewhere in the world are encountered in the Indian subcontinent as well. Several causes, however, are either unique or seen with increased frequency in this part of the world. Compared with industrialized countries, where AKI is mainly a disease of the elderly and is seen primarily in hospitalized patients, community-acquired AKI (CA-AKI) in otherwise healthy young individuals is common in countries in this region, a characteristic they share with other developing countries.³ The median ages of patients with AKI in the higher-income (HICs) and upper middle-income countries (UMICs) are 63 and 64 years, respectively⁴; the average age of patients with AKI in India is 46 years.⁵

Over the years, there has been a change in the spectrum of AKI in the region (Table 77.2).⁶ In urban areas, the causes

Table 77.2 Causes of Community-Acquired Acute Kidney Injury in a Tertiary Referral Center on the Indian Subcontinent (%)

Causes	Number of Patients	Percentage of Total (%)
Infections	294	44
Tropical infections	77	
Acute gastroenteritis	97	
Acute pyelonephritis	31	
Other infections	89	
Obstetric	111	16.6
Sepsis	25	
Postpartum hemorrhage	15	
Preeclampsia/eclampsia	15	
Thrombotic microangiopathy	12	
Others	44	
Envenomation	80	12
Snake	66	
Wasp	11	
Others	3	
Drug-induced acute kidney injury	31	4.6
Poisoning	26	3.89
Paraquat	17	
Others	9	
Rhabdomyolysis	43	6.43
Pancreatitis	56	8.3
Other rare causes	27	4.04
Total	668	

Author's unpublished data.

of AKI are similar to those in the industrialized world, but in rural areas, AKI is mainly the result of diarrheal illnesses, chemicals, snake bites, and insect stings.⁷ Factors responsible for the higher incidence of CA-AKI in these regions include exposure to tropical infections like malaria, leptospirosis, dengue, or acute diarrheal illness; toxic envenomation after snake or insect bites; and the use of unproven traditional or local systems of medicine that frequently include nephrotoxic compounds.⁸ Lethal pesticide poisonings are alarmingly common in India. AKI due to suicidal ingestions of paraquat and rat killer poisons has increased, and that due to copper sulfate is declining.⁹ Paraquat poisoning, which causes AKI, is increasingly reported in the past decade.^{10,11} Sepsis as a cause of AKI has increased almost seven times, from just 1.57% from 1983 through 1995 to 11.43% from 1996 to 2008.¹² The major causes of CA-AKI in the author's hospital have been given in [Table 77.2](#).

Treatment facilities are grossly inadequate. Patients are referred late, often requiring immediate dialysis in the critical care setting. Therefore the mortality rate is high. Intermittent peritoneal dialysis (PD) for AKI is still used in several areas.^{13,14} The high cost of continuous renal replacement therapy restricts its use.¹⁵ The outcome of AKI is worse in the elderly than in the younger population.¹⁶ Conditions causing AKI that are specific to the Indian subcontinent are discussed in the following sections.

PREGNANCY-ASSOCIATED ACUTE KIDNEY INJURY

Improvements in obstetric care have led to the virtual disappearance of pregnancy-related AKI in the industrialized world. Data from a large tertiary care hospital in the eastern part of India have shown that the proportion of obstetric AKI cases remained more or less similar at 11.8% and 12.8% of total cases of AKI over two 13-year time periods (1996–2008 vs. 1983–1995, respectively).¹⁷ A retrospective analysis from Pakistan reported increase in incidence of obstetric AKI in the years 2010–2014 (35.27%) compared with 1990–2010 (21.64%).¹⁸

The frequency distribution of AKI is bimodal in terms of the duration of gestation. The first peak, seen between 8 and 16 weeks' gestation, is associated chiefly with induced septic abortions. The incidence of postabortal AKI has decreased since the legalization and regulation of abortions and the wider availability of medical facilities in the recent times.¹⁹ The second peak in obstetric AKI occurs after 34 weeks' gestation and is related to preeclampsia, eclampsia, abruptio placentae, postpartum hemorrhage, and puerperal sepsis. The incidence of preeclampsia in the Indian population has been reported to be 28.7% in a population-based study.²⁰ Data from Pakistan, Bangladesh, and Nepal have also suggested sepsis related to abortion or puerperium, hemorrhage, hypotension, and preeclampsia as the main associations of pregnancy-related AKI.²¹ The majority of pregnancy-related AKI occurs due to preeclampsia/eclampsia followed by puerperal sepsis and postpartum hemorrhage.¹⁹ Obstetric AKI is the main cause of acute cortical necrosis.^{22,23} Thrombotic microangiopathy is being recognized as one of the important causes of pregnancy-associated acute cortical necrosis.^{24,25} Those women who do not recover kidney function usually undergo contrast-enhanced computed tomography of the kidneys to

look for acute cortical necrosis. The characteristic findings include lack of enhancement of renal cortex, medullary enhancement, and absent renal excretion.²³ If this is absent, a renal biopsy is done in such individuals.

The provision of institutional care and early referral for patients with complicated pregnancies to specialized centers is important to ensure better prevention and management of AKI in such settings. Along with community-level interventions, facility enhancement is also required for decreasing pregnancy-related mortality and morbidity. AKI in pregnancy is managed along the usual lines, with additional cause-specific interventions, if any.²¹ Pregnancy and kidney disease are further discussed in [Chapter 58](#).

ACUTE KIDNEY INJURY DUE TO INFECTIONS

DIARRHEAL DISEASES

AKI secondary to diarrhea is encountered in not only children but also adults. In adults, diarrheal diseases cause 19% of all AKI,¹⁷ whereas 39.5% of CA-AKI in children is caused by diarrhea with dehydration.²⁶ This problem is common in rural areas and urban slums inhabited by poverty-stricken individuals. The incidence increases during summer and the rainy seasons. *Campylobacter*, *Cryptosporidium*, *Norovirus*, *Shigella*, and enterotoxigenic *Escherichia coli* (ETEC) are the most common causes of diarrhea in India.²⁷ Indigenously manufactured rotaviral vaccine is touted to decrease the incidence of rotaviral diarrhea, which is an important of diarrhea in children.²⁸ Consumption of safe and clean water, proper sanitation facilities in every household, adequate nutritional intake by mother and child, safe breastfeeding, proper stool disposal practices, careful case management, and rotavirus vaccination are some of the interventions to be implemented.²⁷

Hemolytic uremic syndrome due to *Shiga* toxin-producing *E. coli* (STEC-HUS) is less commonly observed nowadays.²⁹ STEC-HUS diagnosis is usually made clinically in most centers (history of bloody diarrhea preceding HUS), due to unavailability of *Shiga* toxin assays. Early use of isotonic fluids and appropriate antibiotics are recommended in the management.²⁹ Plasma exchange is used in severe cases

MALARIA

AKI due to malaria predominates in warmer regions closer to the equator in the subcontinent (parts of India, Pakistan, Bangladesh, and Sri Lanka). Most cases of AKI are seen with *Plasmodium vivax* species, whereas severe AKI is most often seen with *Plasmodium falciparum* species.³⁰ Renal endothelial injury is the predominant pathologic finding in kidney biopsies of patients with malaria-induced AKI.³¹ In alignment with the World Health Organization's Global Technical Strategy (GTS) for malaria 2016–2030, India has committed to eliminating malaria by 2030, whereas Sri Lanka was declared malaria free by 2016. Malaria is discussed in further detail in [Chapters 59, 75, and 78](#).

LEPTOSPIROSIS

Leptospirosis, a zoonotic disease, is commonly reported among farmers in India and Sri Lanka. Clinically, it shares common clinical manifestations with another reemerging zoonotic disease causing pulmonary renal syndrome, *Hantaviral* infection. One study from Sri Lanka reported

AKI and hemorrhagic pulmonary syndrome in 82.7% and 62.5% cases of leptospirosis.³² Leptospirosis is discussed in further detail in [Chapter 59](#).

DENGUE FEVER

Dengue fever is the most prevalent mosquito-borne viral disease seen in urban and semiurban areas. Most (70%) of the dengue cases are concentrated in Asia. The majority of complications occur in individuals from 21 to 40 years old.³³ AKI, acute respiratory distress syndrome, and shock associated with dengue carry high mortality.³³ Dengue is discussed in further detail in [Chapter 59](#) and [78](#).

MUCORMYCOSIS

Mucormycosis, previously called zygomycosis, refers to diseases caused by fungi belonging to the order Mucorales. *Rhizopus arrhizus* is the most common etiologic agent of mucormycosis in India. Eleven percent of Indians have diabetes mellitus,³⁴ and 70% of them have poor glycemic control.³⁵ This poses a significant risk factor for development of fungal infections in patients with diabetes. COVID-19 infection led to an increase in the number of mucormycosis cases in India and Pakistan.^{36,37} Organ involvement in mucormycosis occurs through vascular invasion, which leads to thrombosis of large and small arteries and subsequently infarction and necrosis of the affected organ. The major presentations are rhinocerebral, cutaneous, pulmonary, gastrointestinal, and disseminated forms. Renal mucormycosis may occur as isolated renal involvement or as part of disseminated disease and carries high mortality.³⁸ Bilateral renal arterial involvement leads to oliguric AKI. Renal mucormycosis usually develops in otherwise immunocompetent individuals.³⁹ It presents with fever, lumbar pain, pyuria, and oliguria. Computed tomography reveals enlarged, nonenhancing kidneys with perirenal collections and/or intrarenal abscesses ([Fig. 77.1](#)).⁴⁰ The diagnosis can be confirmed by demonstration of hyphae in the material obtained by aspiration or percutaneous biopsy.³⁹ The only definitive treatment is



Fig. 77.1 Renal mucormycosis. This contrast-enhanced computed tomography scan shows bulky kidneys, with large nonenhancing areas (arrows) involving the cortex and medulla and with no contrast in the pelvicalyceal system.

extensive debridement of affected tissue, which may include bilateral nephrectomy, and systemic antifungal therapy with amphotericin B. Bilateral renal mucormycosis carries an extremely poor prognosis.^{38,39}

ACUTE KIDNEY INJURY DUE TO SNAKEBITE AND INSECT STINGS

SNAKEBITE

Snakebite is an occupational hazard and occurs when people are working barefoot in the fields.^{41,42} On the Indian subcontinent, AKI develops after bites by snakes of the viper family.⁴³ In an analysis of 1500 cases from South India, the incidence of AKI after a viperine snakebite varies between 15% and 37.5% depending on the viper species.⁴³

ACUTE KIDNEY INJURY DUE TO SNAKEBITE

Clinical Features

The severity of the symptoms and signs is related to the type of venom, as well as to the dose injected during the bite. Pain, swelling, hemorrhagic blisters, and bruising over the bitten area are commonly seen after hematotoxic snakebites. Cellulitis, necrosis, and gangrene of the bitten areas are seen in up to 20% of the cases after hematotoxic snakebites.⁴³ Capillary leak syndrome associated with Russell's viper venom is characterized by hypotension, hemoconcentration, and generalized edema.⁴⁴ Snake venom metalloproteinases (SVMPs) contribute to capillary leakage by degradation of capillary basement membrane integrity.⁴⁵ Severely envenomed patients experience venom-induced consumptive coagulopathy,⁴⁶ which frequently results in spontaneous bleeding.^{43,47} The number of days taken for the resolution of coagulopathy has a significant association with the development of AKI.⁴³ In majority of the patients with AKI, oliguria often develops rapidly, within the first 24 hours, but it may be delayed until 2 to 3 days after the bite.⁴⁸ Oliguria is the predominant manifestation, though some patients remain non-oliguric.⁴⁸ Hematuria can be seen in 16% of the patients.⁴³

Laboratory investigations reveal varying degrees of anemia and thrombocytopenia, resulting from venom-induced consumptive coagulopathy, thrombotic microangiopathy, and blood loss.⁴⁸ Clotting time is prolonged in 68% of the patients with hemotoxic snakebites. Prolongation of prothrombin time was observed in 98.1% of patients with Russell's viper bites.⁴³ The mean time taken for the coagulopathy to resolve with treatment is 1.5 days.⁴³

Histologic Features

Tubular lesions include acute tubular necrosis and acute interstitial nephritis.^{49,50} Acute tubular necrosis is the predominant lesion seen in 70% to 80% of patients. Bilateral diffuse or patchy cortical necrosis has been observed in 20% of patients with AKI due to snakebite.⁴⁸ Mesangial expansion, mesangiolysis, and mesangial proliferation are the commonly reported glomerular lesions.⁴⁹ The presence of fibrin thrombi in the arterioles is a prominent feature in patients with RCN. A narrow subcapsular rim of cortex often escapes necrosis. The area underlying this rim, however, shows necrosis of glomerular and tubular elements. The

necrotic zone is often bordered by an area of hyperemia and leukocytic infiltration.

Treatment

Early administration of antivenom is vital. Indications for antivenom therapy include prolonged clotting time, spontaneous systemic bleeding, intravascular hemolysis, and local swelling involving more than two segments of the bitten limb. Because only polyvalent antivenom is available in most parts of Asia, precise identification of the snake is not essential for management. Indian studies recommend 10 to 30 vials of polyvalent anti-snake venom.⁵¹ Ten vials of antivenom are administered initially over 1 hour. Each vial of 10 mL antivenom neutralizes 6 mg of Russell's viper venom.⁵² A simple way to monitor antivenom treatment efficacy is by monitoring whole blood clotting time 6 hourly.⁵¹ Repeat doses of 10 vials are required if the clotting time is abnormal after 6 hours.

Other therapeutic measures include replacement of lost blood with fresh blood or plasma, maintenance of electrolyte balance, administration of tetanus immunoglobulin, and treatment of pyogenic infection with antibiotics. Plasma exchange is not recommended in cases of thrombotic microangiopathy due to snakebites.⁵³ The prognosis is good in patients who receive adequate doses of antivenom. Long-term complications include progression to chronic kidney disease (CKD), new-onset hypertension, and hypopituitarism.^{54,55} Mortality due to Russell's viper bites is 13.5%.⁴³

BEE, WASP, AND HORNET STINGS

Honeybees, yellow jackets, hornets, and paper wasps are stinging insects belonging to the order Hymenoptera. An isolated sting causes just a local allergic reaction, but an attack by a swarm of insects introduces a large dose of the venom, sufficient to cause systemic manifestations including AKI.^{56,57} Patients with AKI have been reported to have received from 22 to more than 1000 stings. AKI is secondary to hemolysis and/or rhabdomyolysis.⁵⁸ Hemolysis results from the action of a basic protein fraction, as well as melittin and phospholipase A present in the venom. Rhabdomyolysis has been attributed to polypeptides, histamine, serotonin, and acetylcholine. A direct nephrotoxic role for these venoms has also been suggested. Kidney biopsies show acute tubular necrosis and acute interstitial nephritis. Acute cortical necrosis after a single wasp sting has been reported.⁵⁹ Treatment is usually supportive.

ACUTE KIDNEY INJURY DUE TO CHEMICAL TOXINS

COPPER SULFATE POISONING

Copper sulfate is a strong corrosive that produces symptoms within minutes of ingestion, usually with suicidal intent. Metallic taste, excessive salivation, burning retrosternal and epigastric pain, nausea, and repeated vomiting are the initial features. The vomitus is blue-green in color. Diarrhea, hematemesis, and melena follow. Jaundice, hypotension, convulsions, and coma may develop in severe cases. Acute pancreatitis, myoglobinuria, and methemoglobinemia have also been reported. AKI is seen in 20% to 25% of cases and is invariably oliguric.⁶⁰ Hemoglobinuria may be seen in about 40% of cases. Copper can produce considerable oxidative stress and interferes with the activity of several key enzymes,

such as Na⁺-K⁺-ATPase, G6PD, glutathione reductase, and catalase leading to tubular injury.⁶⁰ Direct nephrotoxicity, severe hemolysis, and hypovolemia secondary to fluid loss are the main factors responsible for kidney injury. Hemoglobin casts may be seen in patients with intravascular hemolysis.^{60,61}

Management includes measures to decrease gastrointestinal absorption, administration of methylene blue for symptomatic meth-hemoglobinemia, and correction of volume deficits.⁶¹ The efficacy of chelation therapy is unproven.⁶¹ Hyperkalemia may be severe and sustained because of the ongoing hemolysis and requires early and frequent dialysis.

ETHYLENE GLYCOL POISONING

Diethylene and polyethylene glycols have been used as cheap substitutes for propylene glycol as vehicles in pediatric syrup preparations. Epidemics of diethylene glycol-induced AKI have been reported in India and Bangladesh.^{62,63} In one large study, 236 deaths were recorded among 339 children with unexplained AKI in a children's hospital in Dhaka, Bangladesh.⁶² Clinical features include liver involvement, hypertension, metabolic acidosis, and AKI.⁶² A recent surge of AKI cases in Gambian and Indonesian children has been attributed to percentage of ethylene glycol in syrups.⁶⁴

ETHYLENE DIBROMIDE POISONING

Ethylene dibromide (EDB), a pesticide fumigant, is absorbed from the skin and intestinal mucosa. Both accidental and suicidal poisonings with EDB have been reported. AKI and hepatocellular injury are the chief manifestations. DIC and pulmonary edema are the other manifestations. Kidneys are involved in 26.4% to 50% of the cases.^{65,66} Early initiation of therapeutic plasma exchange has been shown to remove plasma protein-bound toxin with significant mortality reduction.⁶⁶

PARAQUAT POISONING

Paraquat dichloride is a widely used and highly toxic herbicide. It is one of the common pesticides used to die by suicide in this part of the world.⁶⁷ Accidental poisoning is due to improper handling and usage, as well as lack of personal protective equipment while spraying.⁶⁸ Common symptoms include gastrointestinal corrosive effects with mouth and throat, epigastric pain and dysphagia, acidosis pulmonary edema, shock, arrhythmia, hepatic failure, and AKI. Long-term health effects include pulmonary fibrosis, multiorgan failure, or death. AKI that occurs in 46.4% of patients develops due to the tubular toxicity of reactive oxygen species.⁶⁹ The mortality associated with paraquat poisoning can be as high as 80%.⁶⁸ Steroids, cyclophosphamide, hemoperfusion, and continuous venovenous hemofiltration have shown some mortality benefits; however, larger trials are required.⁷⁰⁻⁷²

HAIR DYE-RELATED ACUTE KIDNEY INJURY

Hair dye and its constituent, paraphenylenediamine, have been reported as an accidental and intentional cause of poisoning on the Indian subcontinent.^{73,74} Acute poisoning with paraphenylenediamine due to accidental or intentional consumption causes edema of the face and neck, resulting in severe respiratory distress that requires

tracheostomy. Microhematuria, cola-colored urine, and oligoanuria are seen in the majority.⁷³ The preceding manifestations are followed by rhabdomyolysis and AKI. In one study, AKI was seen in 32% and RRT requirement was seen in 81.4% of patients.⁷⁴ Mechanisms of AKI are direct nephrotoxicity, myoglobinuria, hemoglobinuria, hypovolemia, and hypotension. Renal lesions are usually found to be glomerular congestion, acute tubular necrosis, myoglobin deposition, and interstitial hemorrhages in patients undergoing biopsy.^{74,73}

ACUTE KIDNEY INJURY DUE TO INTRAVASCULAR HEMOLYSIS AND GLUCOSE-6-PHOSPHATE DEHYDROGENASE DEFICIENCY

G6PD is a key enzyme that protects erythrocytes from oxidative stresses. Deficiency caused by mutations in the *G6PD* gene causes intravascular hemolysis.⁷⁵ The gene is located on the X chromosome. The frequency of G6PD deficiency in the Indian population is 8.5%.⁷⁶ The G6PD variant (Mediterranean) in parts of India and Pakistan leads to hemolysis only in response to oxidative stress.⁷⁷ Individuals deficient in the enzyme cannot maintain an adequate level of reduced glutathione, leading to the precipitation of oxidized hemoglobin in red blood cells, which are then sequestered and lysed. Hemolytic crisis develops within hours of exposure to stress, usually in the form of drugs, certain foods, toxins, or infections. Passage of dark urine followed by oliguria is the most common presentation. Universal newborn screening for G6PD deficiency and avoidance of triggers prevents hemolysis. Adequate hydration with 0.9% normal saline decreases incidence of AKI.⁷⁸ AKI due to mismatched blood transfusions is still reported.⁷⁹

ACUTE CORTICAL NECROSIS

Acute renal cortical necrosis is the most catastrophic of all types of AKI. Of more than 2900 patients with AKI treated with dialysis over 28 years in a study in northern India, 3.8% were found to have acute cortical necrosis.²³ Obstetric complications were responsible for 56% of all cases of acute cortical necrosis, whereas snakebite accounted for 14%. The most striking feature of this condition is prolonged oliguria or anuria. The study in northern India found that only 17% of patients could discontinue dialysis by the end of 3 months.²³

ACUTE PERITONEAL DIALYSIS FOR ACUTE KIDNEY INJURY

PD becomes important in such resource-constrained settings because it requires minimal infrastructure and expertise. When required, and if available, PD is likely to be lifesaving. PD is the preferred modality of choice for AKI in children when surveyed among Indian pediatric nephrologists.⁸⁰ There have been several success stories of acute PD being lifesaving in desperate and emergency circumstances.⁸¹ Both rigid and flexible PD catheters can be inserted at the bedside. The cost of acute PD per day is 10 times lesser when compared with continuous renal replacement therapy.¹⁴ Performance of acute PD is a skill that is easily taught and acquired. In fact, many general physicians, especially in the armed forces, who might be posted to far-off places or places where hemodialysis (HD) is not available around the clock,

routinely carry out acute PD whenever required. During the COVID-19 pandemic in India, because many patients of COVID-19-related AKI did not tolerate intermittent HD, acute PD was preferred by many nephrologists.^{82,83}

CHRONIC KIDNEY DISEASE

INCIDENCE AND PREVALENCE OF CHRONIC KIDNEY DISEASE

An accurate estimate of the number of patients on the Indian subcontinent who have CKD or need RRT is not available owing to the lack of nationwide registries. A countrywide Indian Chronic Kidney Disease cohort study of early-stage CKD patients is ongoing in India.⁸⁴ The study aims to describe the epidemiology of CKD progression and the development of complications such as cardiovascular disease in CKD in developing countries.

A cross-sectional study that screened 5588 adults from different parts of India reported a CKD prevalence of 17.2%, with about 6% having CKD stage 3 or worse.⁸⁵ The prevalence of early stages of CKD (stages 1–3) was reported to be about 13.1% to 15% using the Chronic Kidney Disease Epidemiology Collaboration or Modification of Diet in Renal Disease study equation in a cohort of 3398 otherwise healthy adults.⁸⁶ Studies from Pakistan,⁸⁷ Nepal,⁸⁸ and Bangladesh⁸⁹ have reported a CKD prevalence of 25.3%, 10.7%, and 13.1% to 16%, respectively, in the general population. A systematic review and meta-analysis of population-level prevalence studies in South Asia reported a CKD prevalence of 14% in general population and 27% to 31% in high-risk population.⁹⁰ These figures must be interpreted with caution because of the wide variations in the definition of CKD, study methodology, and sampled population. Few studies have estimated the prevalence of CKD in rural communities.⁹¹ Despite a high prevalence of CKD, there is a low awareness of the disease.⁹²

Commonly used formulas for the estimation of the glomerular filtration rate (GFR), such as the Modification of Diet in Renal Disease study equation, have been validated only in small population-based studies in the Indian subcontinent.⁹³ Differences in body habitus and dietary habits make it likely that these formulas may require further validation and possibly correction factors for the accurate assessment of GFR in this population.⁹⁴ The newer race-independent CKD epidemiologic (2021 Chronic Kidney Disease Epidemiology Collaboration estimated GFR creatinine) equation increases the GFR, lowering the estimated kidney disease burden in the Asian Indian cohort.⁹⁵

The crude and age-adjusted incidences of ESKF in India have been estimated to be 151 and 232/million population, respectively.⁹⁶ This means that 250,000 to 300,000 new patients need kidney replacement therapy every year. Data on the prevalence of ESKF are not available.

DEMOGRAPHICS OF CHRONIC KIDNEY DISEASE AND END-STAGE KIDNEY DISEASE

Diabetic kidney disease, previously restricted to high-income urban residents and older individuals, has now emerged as the most important cause of CKD in this region

Table 77.3 Causes of Chronic Kidney Disease and End-Stage Kidney Failure in India and Pakistan (%)

Cause	India (N = 57,273) ^a	Pakistan (N = 6127) ^b
Diabetic nephropathy	31.3	40.35
Hypertension	12.9	28.89
Chronic glomerulonephritis	13.8	11.57
Chronic interstitial nephritis, calculus disease, obstructive nephropathy	10.4	7.6
Other, unknown	31.8	11.48

^aAdapted from Rajapurkar MM, John GT, Kirpalani AL, et al. What do we know about chronic kidney disease in India: first report of the Indian CKD registry. *BMC Nephrol.* 2012;13:10.

^bData from The Kidney Foundation. Dialysis Registry of Pakistan 2007–2008. Karachi, Pakistan: The Kidney Foundation; 2008.

(Table 77.3). According to the Indian CKD Registry, which has information on more than 57,000 patients, diabetes was listed as the primary diagnosis in 31% of cases of CKD.⁹⁶ In the Indian Chronic Kidney Disease cohort, diabetic kidney disease is the most common cause of CKD (25%), followed by chronic interstitial nephritis and CKD of unknown etiology (CKDu).⁸⁴ In Pakistan, Sri Lanka, and Bangladesh, diabetic nephropathy is an important cause of CKD.⁹⁷⁻⁹⁹ The Indian Council of Medical Research–India Diabetes study reported that 11.4%, 15.3%, and 35.5% of the Indian population have diabetes, prediabetes, and hypertension, respectively.³⁴ The growing incidence of noncommunicable diseases is concerning and is mainly concentrated in the urban population.³⁴ The high incidence of type 2 diabetes in the Indian population is due to economic growth, rapid urbanization, genetic predisposition, increased insulin resistance, poor physical activity, and high refined cereal intake.¹⁰⁰

CKDu is an important cause of CKD in this region and is highly prevalent in certain hotspots.¹⁰¹ Parameswaran and colleagues¹⁰² discovered that CKDu contributes to >50% of the cases of CKD in some cohorts from India. The presentation is usually in the advanced stages. The exact reasons and pathogenesis are unknown; however, the roles of agrochemicals, water quality, heavy metals, and heat stress are implicated.¹⁰³ Predominant tubular involvement without glomerular changes is seen in the kidney biopsy specimens.¹⁰⁴ Chapters 74 and 83 discuss this topic in detail.

Low birth weight and early malnutrition, followed by overnutrition in adult life, have been shown to be associated with the development of metabolic syndrome, diabetes, and diabetic nephropathy in an Indian cohort discussed further in Chapter 20. The finding of a high prevalence of proteinuria and high blood pressure in southern Asian children could be part of this jigsaw puzzle.¹⁰⁵ A high percent of such children can be identified for the first time with CKD.¹⁰⁶ Also not investigated has been the role of dietary habits and indigenous medicines. CKD in individuals with infections such as chronic hepatitis B or C, HIV, and tropical infectious diseases have also been reported.^{107,108}

The mean age of patients with CKD, including those requiring RRT, is generally lower in this region than in other parts of the world.⁹⁶ The mean age of Indian patients with CKD is 50.1 ± 14.6 years and two-thirds are males.⁸⁴ Poorer female representation in the cohorts might reflect a systematic barrier in presentation to health care facilities for females in India, likely due to sociocultural reasons.⁸⁴ A sizable proportion of adults experience rapid GFR decline after diagnosis.¹⁰⁹ Metformin, statins, and renin-angiotensin-aldosterone system blockers are not adequately prescribed in early CKD.¹¹⁰ Usage of tobacco products and poor glycemic control contribute to the high incidence of tuberculosis in the CKD population.^{84,111} India replaced China to be the country with highest number of CKD-related deaths.¹¹² The reasons for such high mortality have not been studied.

CHRONIC KIDNEY DISEASE DUE TO GLOMERULONEPHRITIS

A large study of more than 5400 kidney biopsy specimens at a south Indian tertiary care center that treats not only patients from India but also those from neighboring countries has provided insight into the range of glomerulonephritis in the region.¹¹³ Primary glomerulonephritis was diagnosed in 71% of all the biopsy specimens. Mesangioproliferative glomerulonephritis without IgA was the most common lesion (20.2%), followed by idiopathic focal segmental glomerulosclerosis (FSGS) (17%), minimal change disease (11.6%), membranous nephropathy (MN) (9.8%), IgA nephropathy (8.6%), and membranoproliferative glomerulonephritis (3.7%). Postinfectious glomerulonephritis accounted for 12.3% of all lesions. A report of 3275 biopsies done between 2006 and 2016 from a tertiary care hospital in northern India reported FSGS to be the commonest morphology (18.2%), followed by MCD (16.8%), MN (16%), and IgA nephropathy (10.4%) among the primary glomerulonephritides. Lupus nephritis (10.6%) and amyloidosis (3.7%) were the commonly reported secondary glomerulonephritides.¹¹⁴ MN is more common in adults with nephrotic syndrome >40 years.¹¹⁵ FSGS, MCD, MN, and IgA nephropathy are the common primary glomerulonephritides, and lupus nephritis is the common secondary glomerulonephritis reported from other countries in this region.¹¹⁶⁻¹¹⁸ Secondary amyloidosis is much more common than primary amyloidosis; tuberculosis followed by rheumatoid arthritis are the common etiologies for secondary amyloidosis.^{119,120} The glomerular diseases from the largest published cohorts in the Indian subcontinent are listed in Table 77.4. I-TANGIBLE (Indian TrANslational GlomerulonephrItis BioLogY nETwork) is a newly formed network of investigators in glomerular diseases to collect information in a systematic fashion to understand the clinical outcomes, answer translational research questions better, and identify and recruit patients for clinical trials.¹²¹

CHRONIC KIDNEY DISEASE DUE TO VASCULAR DISORDERS

The most common causes of renovascular hypertension worldwide are fibromuscular dysplasia in the young and atherosclerosis in older adults.¹²² On the Indian

Table 77.4 Spectrum of Glomerular Diseases in Native Kidney Biopsies on the Indian Subcontinent

Country	India ^a	Pakistan ^b	Sri Lanka ^c
Number of patients	3275	1793	2680
Year	2006-2016	1995-2008	2010-2019
Primary glomerulonephritis (%)	73	73	40.9
MCD (%)	14.9	5.8	10.11
FSGS (%)	16.1	21.2	11.41
MN (%)	14.2	17.2	5.7
IgA nephropathy (%)	9.2	1.5	9.14
MPGN (%)	5.1	1.1	3.76
Secondary glomerulonephritis (%)	15.5	10.9	36.9
Lupus nephritis (%)	9.4	4.9	15.45
Amyloidosis (%)	3.3	4.6	0.85
Diabetic nephropathy (%)	1.6	0.9	7.2

^aAdapted from Mittal P, Agarwal SK, Singh G, et al. Spectrum of biopsy-proven renal disease in northern India: a single-centre study. *Nephrology*. 2020;25:55-62.

^bAdapted from Mubarak M, Kazi JI, Naqvi R, Ahmed E, Akhter F, Naqvi SA, et al. Pattern of renal diseases observed in native renal biopsies in adults in a single centre in Pakistan. *Nephrology*. 2011;16:87-92.

^cAdapted from Basnayake DB, Wazil Abdul W, Nanayakkara N, Shafraas M, Priyadarshana P, Mahanama BS. Analysis of a histopathological pattern of renal biopsy specimens in Sri Lanka: a single-centre study. *J Am Soc Nephrol*. 2019.

subcontinent, however, Takayasu arteritis or nonspecific aortoarteritis is the main cause of renovascular hypertension in young adults, as well as in children, accounting for 59% to 80% of all cases in these groups.¹²³ In older adults, as elsewhere, atherosclerosis is the most common cause.¹²⁴

Takayasu arteritis is a granulomatous large-vessel vasculitis (mainly affecting the aorta and its major branches) of unknown cause predominantly affecting young females in their second and third decades.¹²⁵ It was first reported from Japan and has high prevalence among Asians.¹²⁵ The pathology is characterized by the involvement of all arterial layers of the vessel (i.e., panarteritis) with a variable inflammatory infiltrate including acute exudative, chronic, and granulomatous inflammation, situated mainly in the media and adventitia, whereas hyperplasia and neovascularization are observed in the intimal layer. Inflammatory lesions later progress to fibrosis in the arterial wall, resulting in narrowing of the lumen.¹²⁶

Aortoarteritis has been classified into the following types, according to the site of involvement.¹²⁷

- Type I affects the branches of the aortic arch.
- Type IIa affects the ascending aorta, aortic arch, and its branches.
- Type IIb affects the ascending aorta, aortic arch, and its branches and the descending thoracic aorta.
- Type III affects the thoracic aorta, abdominal aorta, and/or renal arteries.
- Type IV affects the abdominal aorta and/or renal arteries.
- Type V is a combination of types IIb and IV.

Involvement of a coronary or pulmonary artery is indicated by appending the suffix *C(+)* or *P(+)* to any of the types.¹²⁸ Type IV is common in India, whereas I and IIa are common in Japan.¹²⁸

The renal artery is narrowed at its ostium and in the proximal third with coexistent stenosis of the perirenal aorta.¹²⁹ Bilateral renal artery stenosis is seen in about 50% of all patients (Fig. 77.2). Nonspecific ischemic

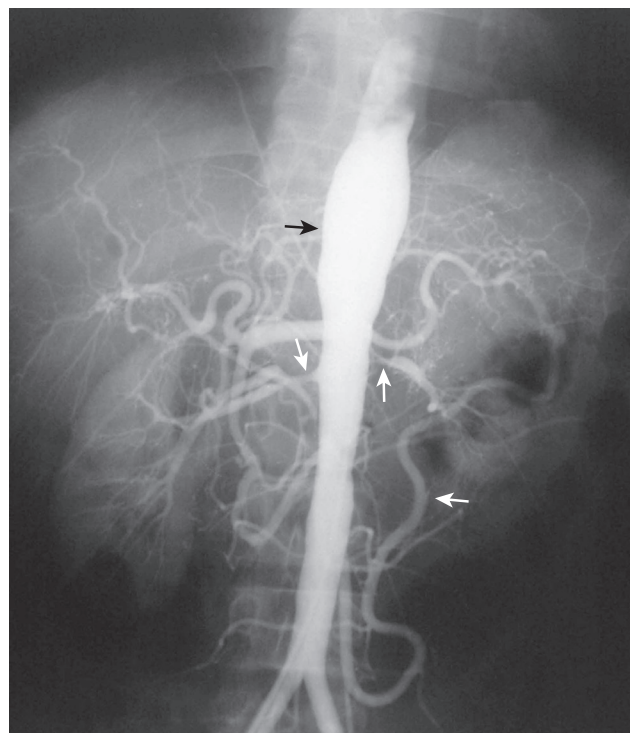


Fig. 77.2 Takayasu arteritis. Shown is bilateral renal artery narrowing (vertical arrows) with poststenotic dilation on the left side, dilation and irregularity of the abdominal aorta (black arrow), and a large collateral vessel (horizontal arrow) arising from the inferior mesenteric artery.

glomerular lesions resulting from arterial narrowing and hypertension are frequently observed in patients with renal artery involvement.¹³⁰ The disease activity is measured by inflammatory markers such as erythrocyte sedimentation rate and C-reactive protein. Vascular imaging like ¹⁸F-fluorodeoxyglucose positron emission tomography/

computed tomography has good sensitivity to detect active vasculitis at diagnosis and during immunosuppression in Indian patients.¹³¹

Oral steroids (prednisolone 0.5–1 mg/kg), along with either mycophenolate mofetil, azathioprine, leflunomide, cyclophosphamide, or methotrexate, are the initial treatment of choice in active disease.^{132,133} Biologics such as anti-tumor necrosis factor- α agents and tocilizumab may be useful in patients refractory to conventional drugs.¹³² Symptomatic vascular lesions in the renal artery may require endovascular stenting and, in rare cases, renal autotransplantation.¹³³

In those who progress to ESKF, creating a vascular access for HD is difficult in the presence of extensive vascular involvement, congestive heart failure, and pulmonary arterial hypertension. The usual cause of death is congestive heart failure.¹³⁰

CHRONIC KIDNEY DISEASE DUE TO RENAL CALCULI

Obstructive nephropathy due to urolithiasis is common in Pakistan and contiguous parts of northern India. This constitutes what has been referred to as a “renal stone belt,”¹³⁴ where nephrolithiasis is responsible for 5% of all cases of ESKD.¹³⁴

The available literature on urinary calculi shows a different stone pattern across the world, highlighting different geographic and causative factors. Studies in northern India have shown that most calculi (>90%) are calcium oxalate stones, predominantly calcium oxalate monohydrate (80%).¹³⁵ Apatite, struvite, and uric acid stones account for <2% each. This incidence of calcium oxalate monohydrate stones is significantly higher than that in Western countries, where such stones constitute up to 50% of the total.¹³⁶ Typically, staghorn calculi are composed of calcium magnesium ammonium phosphate. However, in Indian studies, even staghorn calculi are found to be made up predominantly of calcium oxalate.¹³⁵ The high incidence of nephrolithiasis in India is related to inherited metabolic disorders, dietary factors, and working in high ambient temperatures combined with poor daily water intake.^{137,138} Faith in the efficacy of indigenous medicines to dissolve stones may cause delays in surgical intervention and hastens the development of renal failure.¹³⁹ Nephrolithiasis is discussed in more detail in [Chapter 40](#).

CHRONIC KIDNEY DISEASE DUE TO INDIGENOUS THERAPIES

HERBAL MEDICINE TOXICITY

On the subcontinent, a combination of ignorance, poverty, nonavailability of health care facilities, high cost of modern medicines, and the widespread belief in indigenous systems drives people to turn to indigenous drugs. It is commonly believed that these remedies are gentler and have no side effects. Herbal remedies are often classified as dietary supplements for regulatory and marketing purposes and hence are exempt from rigorous safety testing. However, adulteration of herbal medicines is common in many countries. Moreover, a chemical analysis of such drugs is seldom carried out. The prevalence of Ayurvedic medicines

containing detectable lead, mercury, and/or arsenic was 20.7%.¹⁴⁰ Dwivedi and Dey¹⁴¹ have found high lead and cadmium levels in the leaves of medicinal plants from India.

The medical community has increasingly recognized the potential role of these remedies in causing harm to various organ systems, including the kidneys. Indigenous therapies may cause AKI and CKD.¹⁴² The kidney involvement due to some of the plants used as remedies include chronic interstitial nephritis, renal cysts, renal stones, urothelial malignancies, and renal tubular defects.^{142,143} Prakash and colleagues¹⁴⁴ have described the case of a 60-year-old man who had unexplained kidney failure due to lead toxication from herbal medicines. Mercury is associated with membranous nephropathy.¹⁴⁵ This problem is not confined to the subcontinent; it is relevant to those in the South Asian diaspora.¹⁴⁶

TOXICITY OF NATURAL MEDICINES FROM ANIMAL SOURCES

The raw gallbladder or bile of freshwater and grass carp is used in parts of eastern India to cure bronchial asthma and other ailments.¹⁴⁷ A syndrome of acute hepatic and renal failure has been reported in exposed patients.¹⁴⁷ Symptoms appeared 2 to 12 hours after ingestion and include abdominal pain, nausea, vomiting, and watery diarrhea. Hepatic involvement was observed in 33.3% of patients. AKI sets in within 72 hours and is oliguric in most patients, and urine examination showed microscopic hematuria, proteinuria, and casts (hyaline, granular, and hemoglobin). The duration of AKI ranges from 2 to 3 weeks. Renal histologic examination revealed glomerulitis, acute tubular necrosis, and interstitial edema.¹⁴⁷ Hypotension and hemolysis may also contribute to AKI. Recovery has been universal among patients who have sought medical attention in a timely manner, and death has occurred only in those who reported late and had multiorgan failure.

END-STAGE KIDNEY DISEASE

FINANCIAL AND REIMBURSEMENT ISSUES

Unlike in Western nations, the concept of health insurance (both government funded and private) is still in a primitive stage on the Indian subcontinent. The costs of RRT, therefore, often must be borne by most patients out of their own funds.¹⁴⁸ Some government and private organizations cover the cost of treatment of employees and their dependents as part of employment benefits. The overall cost of RRT in patients with ESKF is less in dollar terms than in the industrialized countries because of the lower staff salaries and low cost of drugs.¹⁴⁹ Nevertheless, it is still several times higher than the current per-capita health care expenditure and remains out of reach for most of the population. In a survey from South India, 91% of patients with HD experienced catastrophic health care expenditure and 77% engaged in distress financing.¹⁵⁰ Provision of medical subsidy is not associated with easing of financial hardship. In a state-funded HD program, 63.5% stopped taking treatment for ESKF (stopped coming to HD centers).¹⁵¹ Costs other than that of HD, like medicines and travel expenses, were not covered. This probably led to higher dropouts and mortality.¹⁵¹ Similar high dropout rates

(18%) were observed at the Sindh Institute of Urology and Transplantation, which provides free dialysis in Pakistan,¹⁵² illustrating that the provision of free dialysis is not enough to overcome access barriers in the region. The cost of dialysis is also influenced by late presentation, with resulting poor clinical status that necessitates hospitalization. Poor hygiene, hot and humid climate, and overcrowding predispose to a variety of life-threatening infections. In one study,¹⁵³ 10.5% of all patients initiated had tuberculosis. The union government of India has launched the Pradhan Mantri National Dialysis Programme, which operates through designated district-level centers to provide dialysis to patients. Apart from this, eligible patients can receive dialysis in other hospitals that are reimbursed under the National Health Protection Scheme (launched in 2018), later renamed Ayushman Bharat Pradhan Mantri Jan Arogya Yojana.¹ HD services have grown using the public-private partnership model under Pradhan Mantri National Dialysis Programme. In this model, governments contract private entities to finance, build, and operate dialysis units at an agreed-upon rate. A public-private partnership offers several advantages, such as economies of scale through strategic purchasing and process efficiencies, scalable training of workforce, opportunities to implement standard operating procedures, and monitoring of quality.¹

HEMODIALYSIS

More than 4950 HD centers existed in India as of 2022, with the majority in the private sector. There were 174,478 patients on maintenance HD as of 2018.¹ Affordable HD through government support is available in 1405 centers in India. Pakistan had 140 dialysis centers in 2004, which increased to 894 in 2023. In both countries, many dialysis units are small, minimal-care facilities, owned and looked after by non-nephrologists or even technicians.¹⁴⁸

More than 50% of CKD patients visited a nephrologist for the first time with CKD stage V.¹⁵⁴ This makes the creation of vascular access for HD difficult, and patients are usually started on temporary dialysis catheters. Decisions on the frequency and duration of HD sessions are based on patient symptomatology, financial considerations, and availability of dialysis slots. Most patients undergo two 4-hour sessions every week. Dialyzer reuse is practically universal, and reprocessing is often performed manually. The absence of regulation by the government or professional societies has prevented standardization of dialysis procedures including the establishment of minimum standards for dialysis machines, water quality, type of dialyzers, and reuse policies.¹⁴⁸

Viral hepatitis is among the most common viral infections encountered in patients undergoing dialysis. The prevalence of hepatitis B and C in dialysis units varies between 2% and 42% and 4% and 45%, respectively.¹ Hepatitis B vaccination, despite low seroconversion rates, has reduced the prevalence among patients in India. Some HD centers still have a high prevalence of hepatitis B infection.¹⁵⁵ After initiation of HD, 0.7% of seronegative patients turn seropositive.¹⁵⁶ Many ESKD patients undergo transplantation with incomplete hepatitis B vaccination schedules and complete the schedules after getting a kidney transplant.¹⁵⁷ Hepatitis C virus (HCV) has emerged as the predominant cause of viral hepatitis in patients undergoing maintenance HD. The annual incidence of HCV infection as detected by anti-HCV antibodies in HD

patients was reported to be as high as 18%, compared with about 2% in patients receiving continuous ambulatory PD.¹⁵⁸ The prevalence of HCV by testing HCV RNA in a study was 8%.¹⁵⁹ In another study, 16.3% seroconverted to HCV RNA positive, within 6 months of initiation of HD.¹⁵⁶ The high incidence of HCV in HD units could be related to the high prevalence of HCV seropositivity in patients undergoing HD, total transfused blood volume, lack of enforcement of universal precautions, and high comorbid illness burden. It has been shown that the isolation of HCV-infected patients during HD significantly decreases the HCV seroconversion rate, from 36.2% to 2.7%.¹⁶⁰

Malnutrition affects most patients receiving dialysis in this region. This leads to a high incidence of tuberculosis.¹⁶¹ Protein-energy wasting is present in 68% to 93% of patients on dialysis from middle and lower socioeconomic strata, and 41% were underweight.¹⁶² The protein and calorie-deficient diet, delay in initiation of dialysis, inadequate dialysis, and infections contribute to this problem. In most of the dialysis units in India, nutritional information is given by health professionals rather than a renal dietician, which is often inadequate. There is also a high prevalence of fatigue, depression, and sleep disturbances.¹⁶³

PERITONEAL DIALYSIS

Facilities for HD are available only in the larger cities. Because PD can be done at home, it could be regarded as the preferred form of dialysis for most patients who live in rural areas and small towns. The quality of life and mortality were better in PD than in HD patients and PD in a study.¹⁶⁴ Despite this advantage, PD continues to be grossly underused with only around 8500 patients on PD in India in 2019.¹⁶⁵ The reason was limited reimbursements for PD compared with state-funded HD; however, the rates have come down with locally produced PD solutions.¹⁶⁶ Other reasons include delayed presentation to dialysis units, which gives insufficient time for patient education and preparation, lack of PD training for nephrologists, and non-PD-centric policies.¹⁶⁶ Concerns are often raised on the grounds that poorly educated patients are likely to be nonadherent to therapy and would be at greater risk of peritonitis owing to the hot humid climate and poor hygienic conditions.

Although PD has been available for almost 25 years in India, at our center this modality is used in less than 10% of patients. Patients are assigned to PD not as a matter of choice but mainly because they have contraindications for other modalities of RRT. In one large hospital, only 8% of PD recipients were started on PD directly; 92% were shifted after receiving HD for a mean duration of about 6 months. Of those, two-thirds were switched because they tolerated HD poorly, 30% because of comorbid conditions and vascular access problems, and 3% because of lifestyle issues.¹⁶⁷ Patients who initially underwent PD were more likely to have diabetes mellitus and coronary artery disease.¹⁶⁷ Their average age was 63 years, compared with 34 years for the HD population. Continuous ambulatory PD is a viable modality of kidney replacement therapy in children in India, where there is lack of access to automated PD and pediatric HD centers. As of 2015, fewer than 250 children in India were on chronic PD.¹⁶⁸ The patient's economic status also determines the PD prescription. The most common practice is to start patients on three 2-L exchanges daily.¹⁶⁹

Cycler-assisted PD and the use of icodextrin solutions remain the exclusive privilege of the rich.

Peritonitis remains the major problem associated with PD on the Indian subcontinent. Peritonitis rates are now around 0.39 to 0.41 episodes/patient-year.¹⁷⁰⁻¹⁷² Unique features of PD-related infections in Indian patients are a high rate of culture-negative cases (65.3%), predominance of gram-negative peritonitis, and increased occurrence of PD peritonitis in the monsoon months.¹⁷³ Gram-negative bacteria constitute 47.8% of all organisms found on culture of patients undergoing PD.¹⁷³ *E. coli* is usually found.¹⁷³ Coagulase-negative staphylococci is the commonest cause of gram-positive PD peritonitis. Fungi (*Candida* spp.) and tuberculosis contribute to 13.3% and 2.2% of the PD peritonitis cases, respectively.¹⁷³ Malnourished patients experienced significantly more peritonitis episodes than patients with normal nutritional status (1.0 vs. 0.2 annually).¹⁷⁴

KIDNEY TRANSPLANTATION

Constraints in operating an effective maintenance dialysis program leave renal transplantation as the only viable option for patients with ESKD.¹⁷⁵ The prevalence of ESKD requiring transplantation in India¹⁷⁶ is estimated to be between 151 and 232 per million population and in Pakistan¹⁷⁷ 100 to 150 per million population. It is estimated that about 220,000 people require kidney transplantation in India. According to the National Organ & Tissue Transplant Organization data, 11,423 kidney transplants happened in 2022. Of these, 86% came from living donors and 14% from deceased donors. Transplantation activity in India still falls woefully short of demand due to lack of finances, lack of an organized cadaver-donor transplant program, and social issues. In transplantations involving living-related donors in India, there is gender bias with predominantly female donors, but the rates of male donation are picking up.¹⁷⁸ In general, kidney transplantation is associated with catastrophic health care expenditures for most families in India.¹⁷⁹ In Pakistan, a free and self-sustaining RRT program has demonstrated 1- and 5-year graft survival rates of 97% and 88%, respectively, which is highly encouraging.¹⁸⁰

Deceased donation rates are low because of absence or ineffectiveness of an organ procurement network, lack of facilities for taking care of potential donors, lack of awareness for donation, religious beliefs and superstitions regarding donation, and lack of faith in the health care system.¹⁸¹ The process depends on the initiative of individual transplant physicians, surgeons, and cooperating intensive care units. Even though more than 150,000 lives are lost annually due to road traffic accidents in India, lack of prompt transport and unavailability of life support services preclude organ donation, in cases where next of kin are willing to provide consent for donating the organs of the deceased. The situation is slowly changing, and deceased donor transplantation is picking up due to public-private partnerships in organ procurement and sharing, employment of transplant coordinators, and establishment of green corridors (route for transportation of organs between hospitals traffic free).

A rise in paired kidney donations and ABO-incompatible transplantations has decreased the waiting time for incompatible donor-recipient pairs.^{182,183} However, altruistic

donation is not allowed in paired kidney donations.¹⁸³ The worldwide shortage of organs for transplantation has given rise to the practice of purchasing kidneys from poor donors by affluent persons. Though officially banned, these kidney trading networks are operational in many south Asian and Middle Eastern countries.¹⁸⁴

Induction immunosuppression agents are either not given or dose down-titrated to bring down costs and prevent infections.¹⁸⁵ Antithymocyte globulin is the commonly used induction agent. Steroid withdrawal is scarcely practiced. Mycophenolate mofetil (MMF) with tacrolimus and steroids are the maintenance drugs commonly prescribed. Studies depicting long-term outcomes are few.¹⁸⁶ Infections and nonadherence are the common causes of mortality in children.¹⁸⁶ Tuberculosis in a posttransplant setting is common and affects 5.7% to 12.3% of renal transplant recipients in southern Asia.¹⁸⁷ It results from reactivation of a quiescent focus and manifests in the first year after transplantation in more than 50% of patients. Although pleuropulmonary involvement is the most common, disseminated disease (with lymph node involvement) occurs in about 30% of patients. Unusual sites of involvement include the skin, tonsils, vocal cords, and prostate. Renal transplant recipients with tuberculosis present numerous diagnostic difficulties.¹⁸⁷ The Mantoux test is generally unhelpful. Classic radiologic findings are seen only in a minority, examination of a sputum smear for acid-fast bacilli has a low yield, and culture takes 4 to 6 weeks. Bronchoalveolar lavage, bone marrow biopsy, and liver biopsy must be used to make a diagnosis of tuberculosis.

There are also problems with the treatment of tuberculosis in transplant recipients, specifically in selecting antituberculous drugs and determining the duration of therapy.¹⁸⁷ Rifampicin is a well-known hepatic P450 microsomal enzyme inducer that increases the clearance of both prednisolone and calcineurin inhibitors. The dosage of calcineurin inhibitors needs to be increased to maintain therapeutic blood levels. This change raises the cost of therapy and is unacceptable to most patients. An alternative regimen that has been successfully used consists of a combination of isoniazid, ethambutol, pyrazinamide, and fluoroquinolones.¹⁸⁷ The optimal duration of therapy is a matter of debate. For combinations using rifampicin and isoniazid, 12 months of treatment have been recommended.¹⁸⁷ However, the duration is increased to 18 months for patients who are not receiving rifampicin.¹⁸⁸ Such patients generally receive two drugs, usually a fluoroquinolone and isoniazid for 18 months. The initiation of isoniazid chemoprophylaxis during dialysis prevents the development of tuberculosis after transplantation.¹⁸⁹ However, isoniazid can cause hepatic dysfunction, for which it often must be discontinued.¹⁸⁹ Drug-resistant mycobacterial strains are also of concern, and the incidence of primary isoniazid resistance has been increasing steadily.¹⁸⁷ The role of isoniazid prophylaxis thus remains controversial.¹⁸⁷

Other common infections in transplant recipients are hepatitis B and hepatitis C, both of which are highly prevalent in patients undergoing dialysis. Hepatitis B is encountered in about 5%, whereas hepatitis C is seen in 15% to 20%. With the availability of direct-acting antiviral drugs,¹⁹⁰ most patients now receive treatment that they were not getting during the interferon era due to cost and side

effects. Patients now enter sustained remission with these drugs. Although patient and graft survivals at 5 years were reported to be similar in HCV-positive and HCV-negative patients, serious bacterial infections were significantly more common in HCV-positive patients.¹⁹¹ The rate of cytomegalovirus infection was found to increase from 4% in the years before calcineurin inhibitors were widely used to 14.5% in the calcineurin era.¹⁹² Primary infection is seldom seen, however, because the vast majority of both donors and recipients are seropositive.¹⁹² Compared with the western data (where invasive candidiasis is the commonest etiology), mucormycosis (27%), invasive aspergillosis (23%), and *Pneumocystis jirovecii* are the common fungal infections in kidney transplant recipients.^{193,194}

SUMMARY

The spectrum of both acute kidney disease and CKD on the Indian subcontinent differs significantly from that encountered in the industrialized world and other developing countries. Community-acquired AKI in otherwise healthy individuals continues to be common in the region. Although glomerulonephritis has been the most common cause of ESKF, it is being replaced by diabetic nephropathy. The lack of an adequate number of nephrologists and the high cost of treatment pose the biggest challenges to the management. Infections are an important cause of morbidity and mortality in patients with kidney disease in this region.



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QUESTIONS

1. A 30-year-old previously healthy man has come to the emergency department of a community health center with complaints of loose motions, nausea, vomiting, and fever for 3 days and decreased urine output for 2 days. He is dehydrated and in shock. After initial resuscitation with intravenous fluids and other symptomatic treatment, his blood pressure has risen to 120/80 mm Hg. His investigations reveal serum creatinine of 4.6 mg% and serum potassium of 6.9 mEq/L. Even after 6 hours of treatment, he has not produced any urine and serum potassium is still 6.5 mEq/L. His transportation to a next higher facility will take another 12 hours. Acute peritoneal dialysis over a stiff catheter is possible in the center. Which of the following is most appropriate in these circumstances?
 - a. Shift the patient immediately to a higher center, where facilities for hemodialysis are available.
 - b. Start acute peritoneal dialysis over a stiff catheter and shift when the patient becomes stable.
 - c. Continue with aggressive fluid administration and similar management for another 6 hours and decide later.

Answer: b

Rationale: The patient most likely has cortical necrosis. The diagnosis can be made with contrast computed tomography. If the computed tomography does not suggest cortical necrosis, a biopsy may be indicated.

2. A 26-year-old woman suffered massive postpartum hemorrhage after childbirth and remained in shock for 6 hours. Subsequently, she developed acute kidney injury (AKI), and her serum creatinine increased to 7 mg%. She has been on hemodialysis for the past 3 weeks, and her 24-hour urine output is 200 mL. Her urinalysis has

shown trace albuminuria and occasional red blood cells, without any casts. What is the most appropriate next step in management?

- a. Perform a renal biopsy.
- b. Do a contrast-enhanced computed tomography scan of the kidneys.
- c. Declare her as suffering end-stage kidney disease and plan for renal transplantation.
- d. Start her on steroids and plasmapheresis.

Answer: b

Rationale: The patient most likely has cortical necrosis. The diagnosis can be made with contrast computed tomography. If the computed tomography does not suggest cortical necrosis, a biopsy may be indicated.

3. In which of the following circumstances would you make a diagnosis of chronic kidney disease (CKD) of undetermined cause?
 - a. CKD with unilaterally small kidney and hypertension, without evidence of renovascular insufficiency
 - b. CKD in a patient with diabetes of long duration in whom a renal biopsy has not been done to confirm diagnosis
 - c. CKD with symmetrically small kidneys and no albuminuria, without any risk factors
 - d. CKD in a patient with a preceding history of edema, nephrotic-range albuminuria, and hypertension for 3 years but no renal biopsy diagnosis

Answer: c

Rationale: The pathophysiology of CKDu is generally a tubulointerstitial process leading to small kidneys and is associated with absence of proteinuria in the absence of other common risk factors.